

Bluestock Mutual Fund Analytics & Performance Intelligence

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Institute: Mangalmay Institute of Engineering and Technology

Project: Mutual Fund Analytics

Technology: Python | SQL | Power BI

Academic Year: 2026

Executive Summary

Bluestock Fintech is a financial technology company focused on democratising investment analytics for retail and institutional investors in India. This capstone project tasks you with building a full-stack Mutual Fund Analytics Platform that ingests publicly available data from AMFI India, transforms it through a robust ETL pipeline, stores it in a relational database, and presents insights via an interactive dashboard.

Problem Statement

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors often struggle to make data-driven fund selection decisions due to fragmented data, lack of unified analytics platforms, and complex financial metrics. This project solves real business problems:

P1: Data Fragmentation

Problem	NAV data, AUM data, SIP flow data and portfolio holdings are available on different sections of the AMFI website in different formats (TXT, PDF, HTML tables). No single unified database exists.
Solution	Build an ETL pipeline that consolidates all data into a single SQLite/PostgreSQL database.

P2: Performance Comparison Gap

Problem	Investors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe ratio, Alpha, Beta, etc.
Solution	Compute and visualise all key risk-return metrics in a single dashboard.

P3: No Benchmark Tracking

Problem	Most retail investors do not know whether their fund is outperforming its benchmark index. Tracking error and information ratio require both NAV and benchmark index data.
Solution	Join NAV data with benchmark index prices and compute rolling alpha and tracking error.

P4: Investor Behaviour Blind Spot

Problem	AMCs and distributors have limited visibility into how demographic and geographic factors influence SIP amounts, fund preferences and redemption patterns.
Solution	Analyse investor transaction data to generate demographic segmentation and behavioural insights.

P5: Slow Reporting

Problem	Monthly MF reports are static PDFs that take days to prepare. Stakeholders need live, self-service dashboards with drill-down capability.
Solution	Build a Power BI / Tableau dashboard refreshable from the ETL pipeline output.

Project Objectives

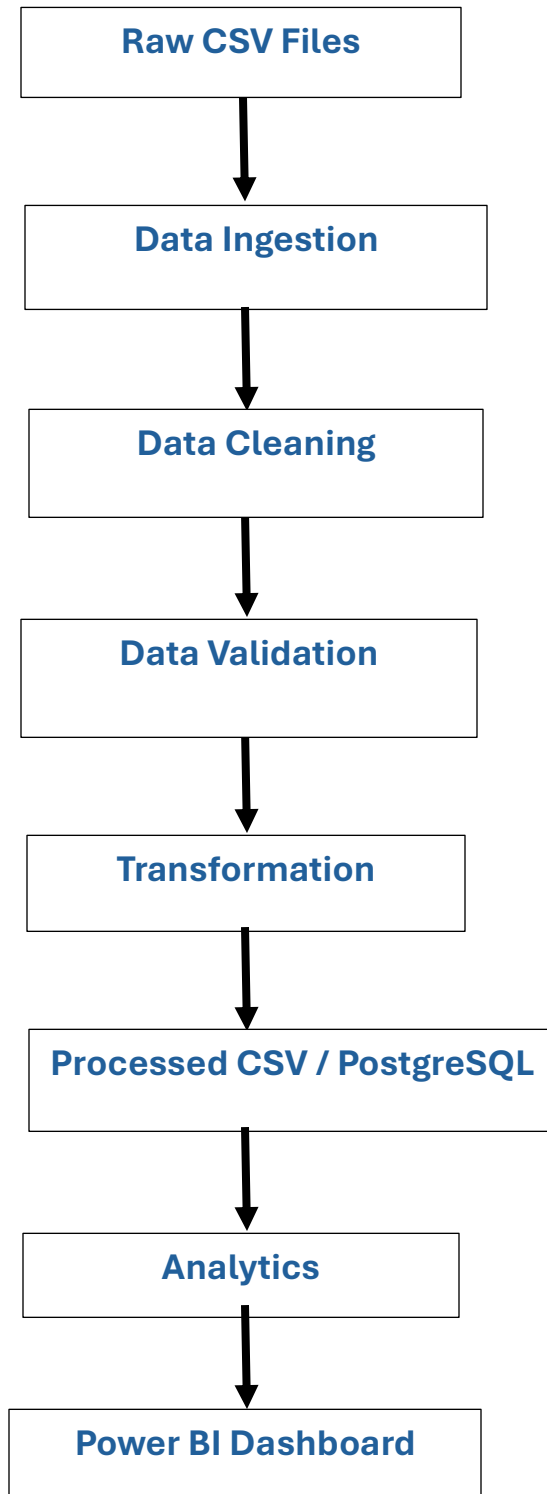
#	Objective	Outcome
O1	Build an ETL pipeline from raw AMFI data	Automated Python script
O2	Design a normalised SQL schema for MF data	5-table star schema
O3	Perform comprehensive EDA on NAV & AUM data	EDA notebook with charts
O4	Compute performance & risk metrics per scheme	Metrics dashboard
O5	Build an interactive BI dashboard	Power BI / Tableau file
O6	Analyse investor transaction patterns	Demographic insights
O7	Compare fund returns vs benchmark indices	Alpha / tracking error report
O8	Document and present the entire project	PDF report + slides

Data Sources

All datasets used in this project are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

Dataset	Purpose
01_fund_master.csv	Fund identification and classification
02_nav_history.csv	Historical NAV analysis
03_aum_by_fund_house.csv	AMC/AUM analysis
04_monthly_sip_inflows.csv	SIP trend analysis
05_category_inflows.csv	Category-level inflow analysis
06_industry_folio_count.csv	Folio growth analysis
07_scheme_performance.csv	Return/risk analysis
08_investor_transactions.csv	Investor behavior
09_portfolio_holdings.csv	Portfolio/sector analysis
10_benchmark_indices.csv	Benchmark comparison

ETL Architecture



ETL Design

An ETL pipeline was developed to convert raw mutual fund data into clean, analysis-ready data.

Extract: CSV datasets containing fund master, NAV, AUM, SIP inflows, transactions, benchmarks, and performance data were loaded using **Python and Pandas**.

Transform: Data was cleaned by standardizing column names, converting dates and numerical fields, handling missing values, removing duplicates, validating AMFI codes, standardizing categories, and preparing performance metrics such as return, risk, alpha, beta, and Sharpe ratio.

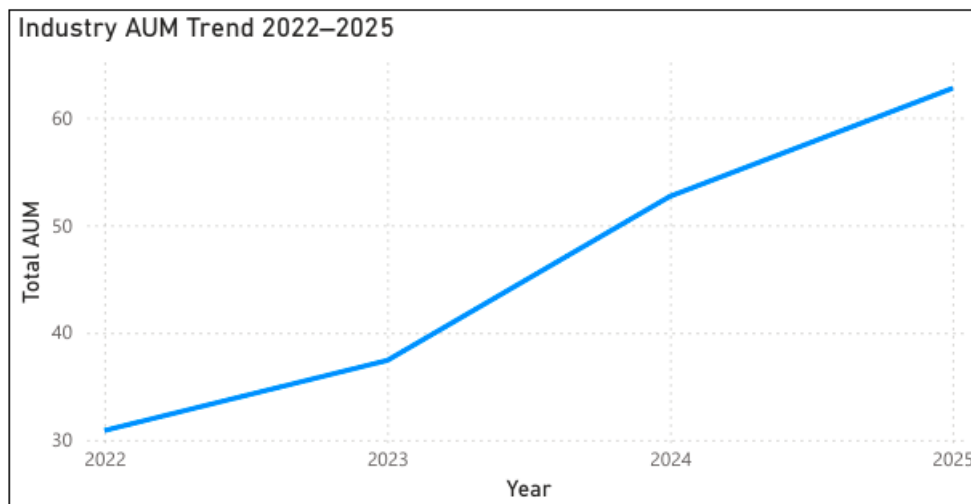
Load: The processed data was stored as cleaned datasets/database tables and connected to **Power BI** for the four-page interactive dashboard.

Overall, the ETL process improved **data quality, consistency, and reliability** for analysis and visualization.

Industry Analysis

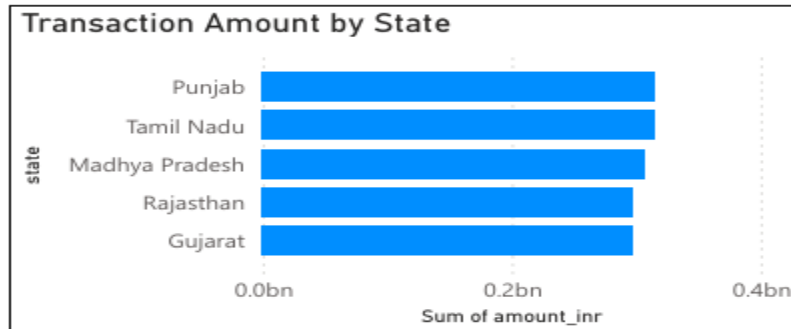
The Industry Analysis demonstrates the overall growth and scale of the mutual fund ecosystem. AUM shows an increasing trend between 2022 and 2025, while major AMC's such as SBI Mutual Fund, ICICI Prudential MF and HDFC Mutual Fund represent a significant portion of the analysed fund-house universe. Strong SIP inflows and 26.12 crore investor folios further indicate substantial investor participation. Overall, the analysis highlights a growing and competitive mutual fund industry, where AUM expansion, systematic investment participation and AMC-level market presence are key indicators of industry performance.

1. AUM trend

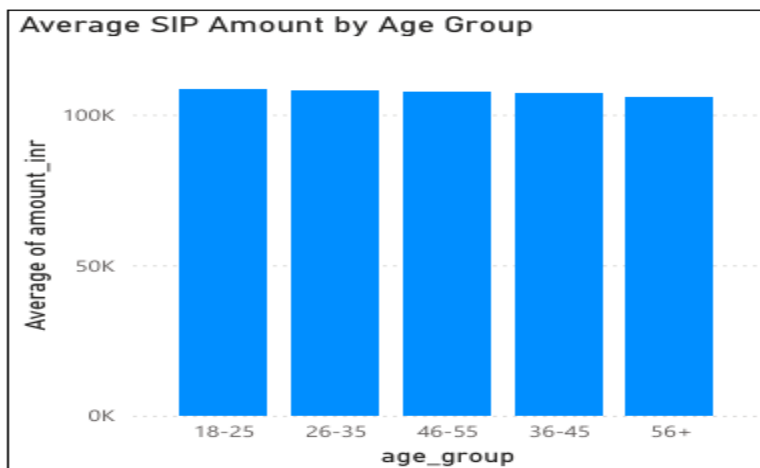


Investor Analytics

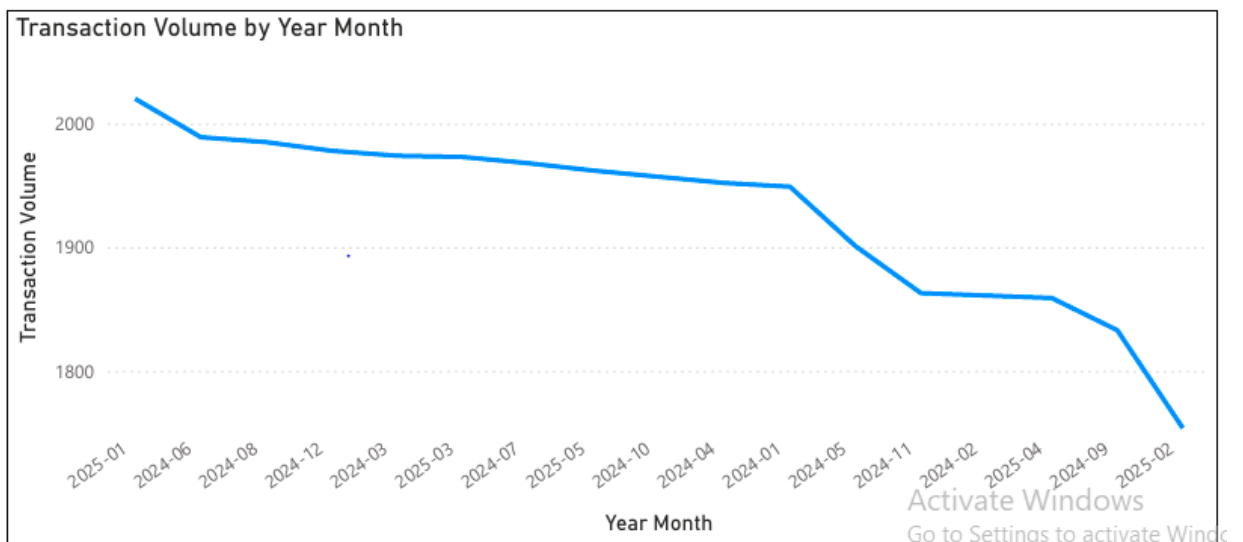
1. Transaction amount by state



2. Age-group analysis

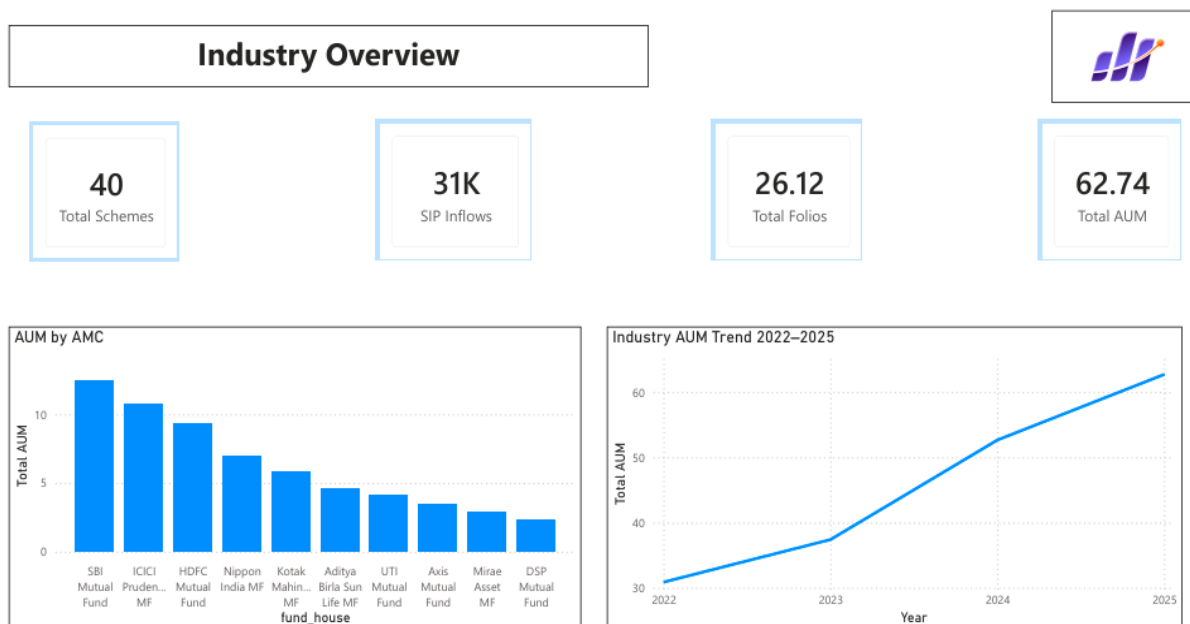


3. Monthly transaction volume



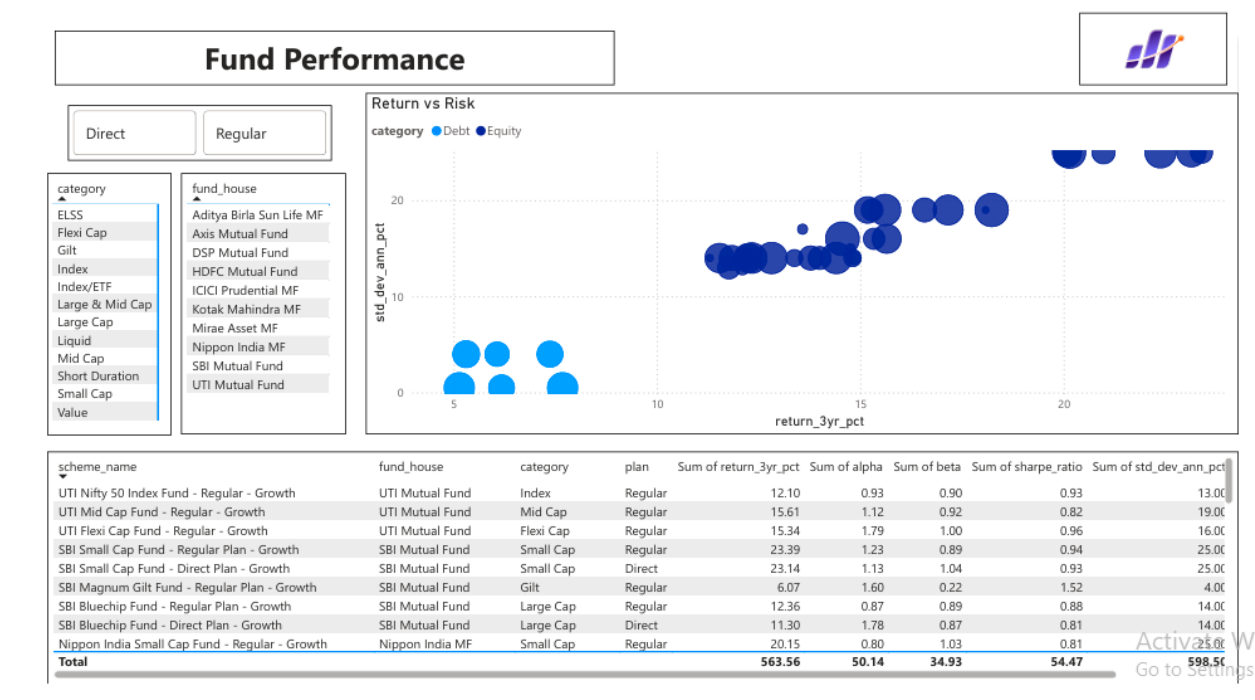
Industry Overview Dashboard

The Industry Overview dashboard indicates a strong mutual-fund ecosystem characterised by substantial AUM, significant SIP participation and a large investor folio base. The AUM comparison also highlights the dominance of established AMC's such as SBI, ICICI Prudential and HDFC. Overall, the dashboard provides management with a clear view of industry scale, AUM growth and AMC-level concentration, supporting informed monitoring of market structure and fund-house performance.



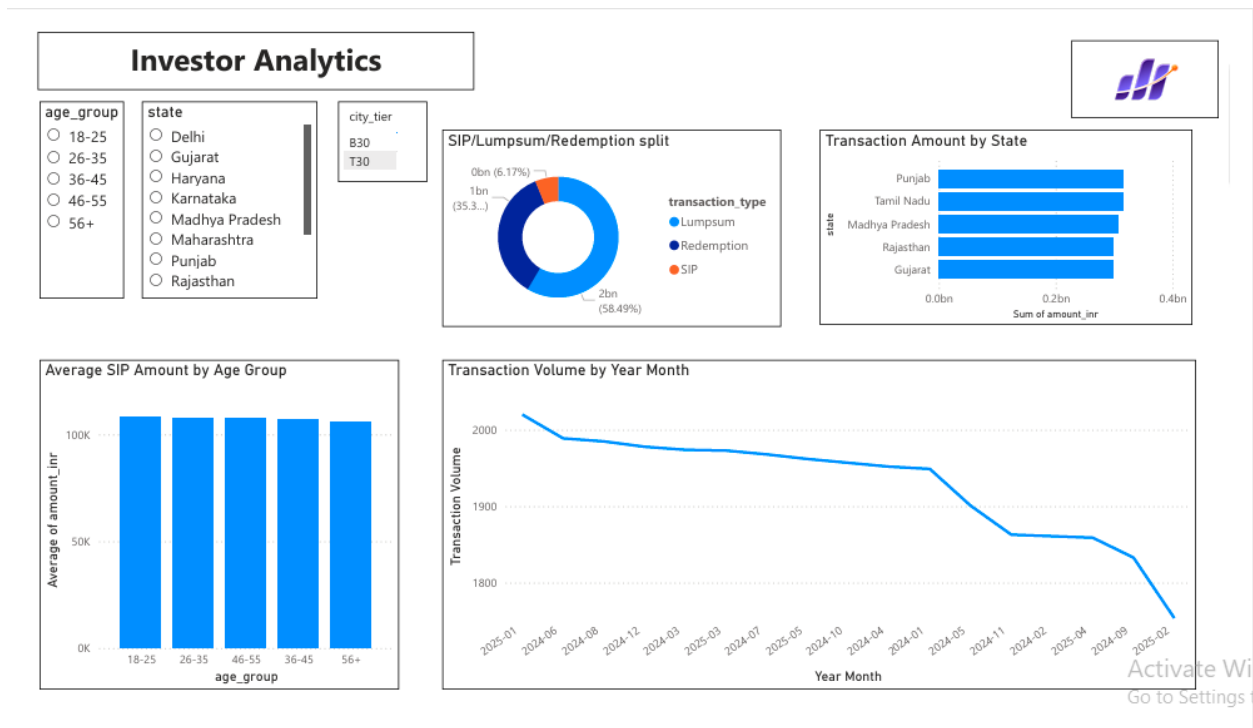
Fund Performance Dashboard

The Fund Performance dashboard highlights a clear relationship between return and risk across mutual fund schemes. Higher-return small-cap funds generally exhibit greater volatility, while debt-oriented funds such as SBI Magnum Gilt Fund demonstrate lower volatility and strong risk-adjusted performance. The inclusion of Alpha, Beta, Sharpe ratio and standard deviation provides a more comprehensive basis for fund comparison than return alone. Overall, the dashboard enables investors and analysts to evaluate funds according to both performance and associated risk.



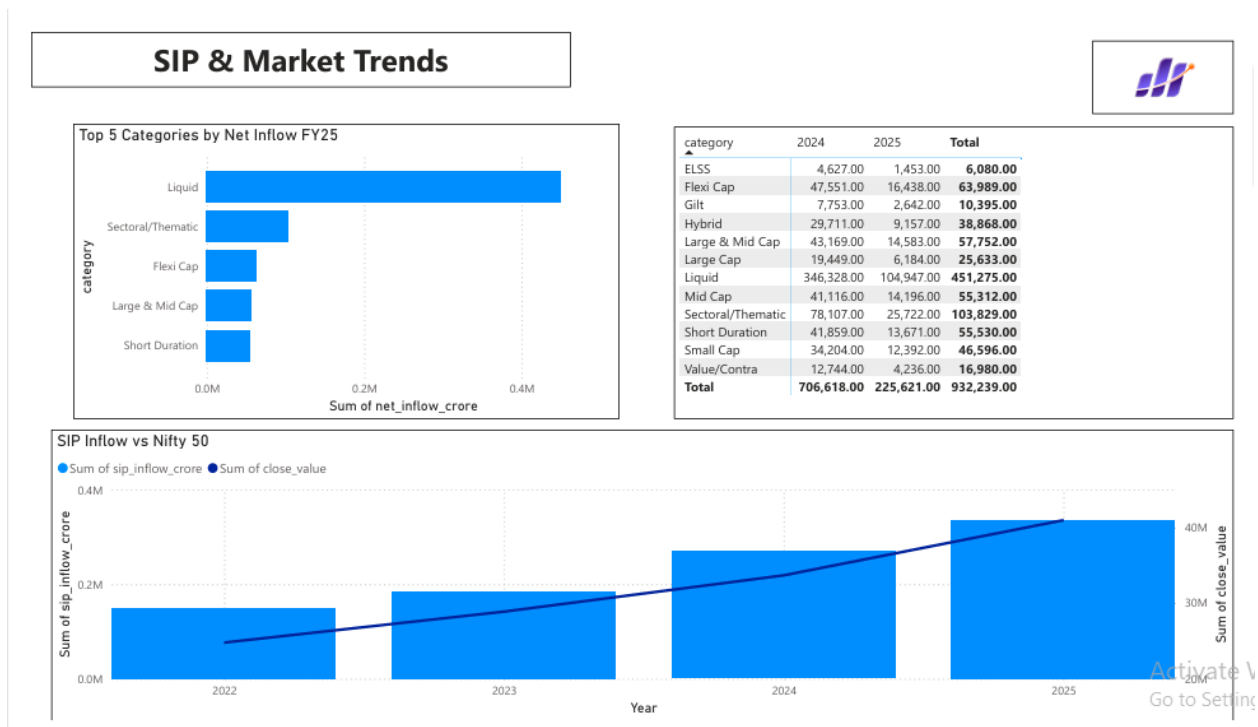
Investor Analytics Dashboard

The Investor Analytics dashboard provides a comprehensive view of investor behaviour by analysing transaction amounts across states, transaction types, age groups and city tiers. The analysis demonstrates variations in investment activity and average SIP amounts across demographic segments, while the monthly transaction trend highlights changes in investor participation over time. The dashboard can support segmentation and behavioural analysis; however, because the underlying transaction data is synthetic, these findings should be considered illustrative rather than representative of actual investor behaviour.



SIP & Market Dashboard

The SIP & Market Trends dashboard demonstrates the growth and distribution of SIP and category-level investment flows while comparing SIP activity with Nifty 50 market performance. Liquid, Sectoral/Thematic, Flexi Cap, Large & Mid Cap and Short Duration emerge as key categories in the displayed net-inflow analysis. The combined SIP and market view provides useful context for understanding investor-flow trends alongside market movements and highlights the growing importance of systematic investing in the mutual fund ecosystem.



Key Findings

- **Strong Growth in Mutual Fund Industry:** The analysis shows an increasing trend in AUM from 2022 to 2025, indicating expansion of the mutual fund industry and growing assets under management.
- **Significant Investor Participation Through SIPs:** SIP inflows are a major component of the industry. The project records ₹31,002 crore of SIP inflow in December 2025, demonstrating strong participation through systematic investing.
- **Major AMCs Have Strong Market Presence:** The AMC analysis highlights major fund houses such as SBI Mutual Fund, ICICI Prudential MF, HDFC Mutual Fund and Nippon India MF as prominent participants in the analysed dataset.
- **Fund Performance Involves a Clear Risk–Return Trade-off:** The Fund Performance dashboard shows that funds with higher returns can also have higher volatility. For example, SBI Small Cap Fund has a 23.39% three-year return with 25% standard deviation, while SBI Magnum Gilt Fund has a 6.07% return with only 4% standard deviation.
- **Investor and Category Behaviour Varies Significantly:** The Investor Analytics and SIP dashboards reveal differences in transaction activity across states, age groups, transaction types and fund categories. The SIP analysis identifies Liquid, Sectoral/Thematic, Flexi Cap, Large & Mid Cap and Short Duration among the leading categories by displayed net inflows.

Recommendations

- **Automate data collection** to improve accuracy and reduce manual work.
- **Improve data validation** by checking missing and duplicate records.
- **Use risk-adjusted metrics** along with returns for fund evaluation.
- **Strengthen investor segmentation** using age, state and transaction type.
- **Enhance the Power BI dashboard** with advanced analytics and automated refresh.